

What a Projection Fold Can and Cannot Imitate

The Complete Arc of a Campaign’s Founding Question, from a Kinematic Theorem to a Sealed Analyticity Exponent

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A smooth worldline in a hidden manifold, observed through a projection that folds with respect to a detector’s time, produces two apparent branches of opposite orientation where there was one. That is a theorem about maps. Whether any local hidden dynamics makes such folds at a rate with the structure physical pair production has is not a theorem, and this paper reports the whole attempt to find out, across twenty-nine runs on declared finite models, with every failure kept. The instrument net reproduces the exact controls and refuses the traps, a degenerate critical point is never classified as a fold, and a thousand perturbed members of the negative control each produce exactly twelve folds, which is what a rate set by the initial-condition measure looks like. In the one populated family the campaign could seal, suppression of time reversal is organized by a Gaussian tail in a measured effective gap and the Schwinger-shaped variable fails to compete, the first predicting held-out cells at 0.38 of its own training error and the second at 3.14 times the first. The family built to escape that verdict turned out to contain no events at all, empty at every one of twelve probed cells, because its alternating slabs cancel between 95 and 99 percent of every deposit with no fixed sign. Both mandatory gates then passed their bars on grids that contained none of the events their claims were about, and both had to be rerun on probe-placed cells, where conservation holds with every census summing exactly to twenty thousand and the orientation charge rule failing zero times. The model carries no quantum phase structure. A two-level mode swept twice through an avoided crossing interferes at a level no noise floor approaches, while the matched classical ensemble carries only the smooth trend its own pulse overlap predicts, its residual sitting in the middle of its null. Reaching that statement took four corrections, each to an error of the declaration rather than of the model. A same-species decay of one particle into two is excluded by parity rather than missed, since across 441 generic levels on four folding worldlines the observed branch count changed by minus two, zero, or plus two and never by anything else. Declaring one branch class invisible produced 63 apparent decays with missing energy whose counts the consumer’s own detector model predicted exactly, so apparent decay here measures what a consumer cannot see. Lifetimes in this family are not memoryless, their coefficient of variation measuring 0.159 against the value one an exponential requires. The founding question is then answered under seal for one observable. A deterministic crossing of a declared field profile is exponentially suppressed with an exponent fixed by the distance to the field’s nearest complex singularity, four fitted values agreeing within 1.07 percent across widths spanning a factor of 2.56, with no distribution anywhere whose tail could be responsible. The observable is a per-crossing transfer and not an event rate, and nothing here is evidence that physical pair creation or decay works this way.

I. INTRODUCTION

Take a curve in some space of hidden states and look at it only through a map into spacetime. A detector picks a time function on spacetime, and the observed history of the curve is the sequence of places where it meets each level of that time function. If the time function increases along the curve everywhere, the observer sees one particle moving forward. If it has a nondegenerate critical point, the observer sees one branch turn into three, or none into two, at a single instant, and the two new branches run in opposite directions of the detector’s time. Under the standard identification of the direction of proper time with the sign of charge, that is an apparent pair event [3, 4].

This is where the campaign that produced this paper started, and it separates three claims that must never be run together. The first is kinematic. A nondegenerate fold of the observed time along a smooth worldline produces exactly two extra branches of opposite orientation, and the signed count is unchanged. This is a theorem about smooth maps and it holds wherever its smoothness hypotheses hold [1, 2]. The second is dynamical and holds only inside a declared model. It asserts that some specified local hidden dynamics produces such folds at a stable rate that is not put in by hand. The third is physical. It asserts that the rate and the observables reproduce quantum field theory, including the exponential suppression Sauter and Schwinger computed for a constant field [7–10]. A successful demonstration of the first is not evidence for the second, and a successful demonstration of the second in one declared family is not evidence for the third. Every section below says which of the three it is about.

An anti-circularity contract was written before any

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run and governs all of them. No candidate model may contain a command to fold at a wanted rate, may take the Schwinger exponent as an input probability, may be fitted to a target rate on the range it is then evaluated on, may postselect the trajectories that fold or escape or resemble particles, may use observer-specific coordinates that are not transformed along with the observer, or may assign energy and charge after inspecting which branch pairing would be convenient. What a model may contain is local fields, a metric or an action, a projection, and a source distribution fixed before the claim-bearing sweep. Claims are carried only by sealed preregistrations whose bars, grids, estimators, and stopping rules are fixed in advance and whose file hashes are recorded in a seal ledger before the governed runner executes, which is the discipline the whole campaign runs under [21–23].

The arc has one further discipline that it did not start with and had to earn. Five separate runs in this arc passed or reported on manifests that contained none of the events their claims were about, and each time the error was in the manifest rather than in the instrument. The rule that came out of them is now unconditional in this campaign. Every registration must cite a committed probe verifying that its manifest contains the variation its claim is about, for every bound cell and every bound member, and any arm of any experiment must do the same even when the arm is a new construction rather than an inherited grid. Section X reports all five as record facts and names the design error in each, together with four bars that failed for reasons belonging to the declaration rather than to the model.

The scope of the whole paper is fixed here and repeated in the ledger of Sec. XI. Every dynamical statement below is measured in a declared finite model with declared constants, integrated numerically at a declared step, and it is labeled as such. Two prior papers of this campaign carry parts of this arc in full detail and are cited rather than restated, one on the replication of a published classical pair event [19] and one on the sealed Schwinger-scaling negative [20]. No result reported here is evidence that physical pair creation or physical particle decay works this way, no claim about quantum field theory is made or implied, and the one sealed positive result of the arc is about a per-crossing energy transfer in a declared one-dimensional model and not about an event rate.

II. THE INSTRUMENT NET AND THE NEGATIVE CONTROL

Let the hidden state be $z(\tau)$ on a smooth manifold, let π carry it to spacetime, and let a detector declare the time functional $T_u(x) = u_\mu x^\mu$. An apparent pair event is a nondegenerate critical point of the observed time,

$$\frac{d}{d\tau}T_u(x(\tau)) = 0, \quad \frac{d^2}{d\tau^2}T_u(x(\tau)) \neq 0, \quad (1)$$

near which the observed time has the normal form $T_u - t_0 = \sigma(\tau - \tau_0)^2$ with $\sigma = \pm 1$. Crossing such a point, the unsigned number of preimages of a constant-time slice changes by exactly two while the signed number, counting each preimage with the sign of $dT_u/d\tau$, does not change at all. That is the kinematic claim, and it is the only statement in this paper with the label [proved].

The instrument that finds these points was validated against exact controls before any dynamics ran, and the tolerances were then frozen and sealed. The controls are a monotone null, an exact creation fold, an exact annihilation fold, a degenerate cubic critical point, and a double fold, in that order the cases the classifier must find, must find with the opposite orientation, must refuse, and must find twice [1]. On the monotone null the instrument reports one branch and zero critical points at every slice. On the creation fold the branch count runs 0, 1, 2 while the signed count stays 0 at every slice, and on the annihilation fold it runs the other way. The cubic control produces one critical point which is classified degenerate and is never counted as a fold. The double fold $t = \tau^3 - \tau$ produces band counts 1, 3, 3, 3, 1 with signed count +1 everywhere and both critical points located at $\pm 1/\sqrt{3}$ with the correct types. The fitted separation exponent of the square-root law agrees with 1/2 to better than 10^{-12} , and the two independent implementations agree on branch locations to 4.5×10^{-16} .

The classifier constants are derived from those margins rather than chosen. The first-derivative tolerance sits at 10^{-9} against a worst measured residual of 4.6×10^{-13} at located events, and the second-derivative tolerance sits at 10^{-8} against the smallest genuine fold curvature of 0.208 found anywhere in the committed evidence, so no regular point can be read as critical and no fold and degenerate point can be exchanged. The freeze states its own invalidation condition, that a claim-bearing model whose fold curvatures approach 10^{-5} in campaign units forces a re-freeze before use.

Structural stability was measured next, because a fold that does not survive perturbation is a coincidence rather than a phenomenon. The exact fold was perturbed by bounded random smooth terms at twelve perturbation norms from 10^{-5} to 0.316, with 500 independent trials at each. Fold persistence is exactly 1 at every norm up to 0.123 and falls to 0.956 at the largest, where 22 of 500 trials lose the fold, and no trial at any norm produced a degenerate classification. The median fitted separation exponent stays at 0.5 across the whole sweep, departing from it by at most 2.8×10^{-6} at the largest perturbation. Simple folds persist, which is what singularity theory predicts and what makes the rest of the arc worth running.

The negative control is the campaign's most useful early measurement. A bounded local Hamiltonian with an oscillator in the observed time, an anharmonic hidden coordinate, and a bilinear coupling produces folds readily, and the folds are rigid. A single member integrated by velocity Verlet at step 10^{-3} produces exactly 12 folds, with 7 positively oriented and 6 negatively oriented

segments and a relative energy drift of 1.3×10^{-7} . A thousand-member ensemble with perturbed initial conditions produces exactly 12 folds in every single member, 12000 folds in total with zero dispersion, at a mean relative drift of 6.7×10^{-10} . The same scenario run on a second machine in a container reproduces the outcome hash bit for bit.

That uniformity is the point. A rate is a property of the initial-condition measure and the dynamics together, and in a bounded Hamiltonian with a smooth measure there is nothing to make it exponentially small. Exponentially suppressed pair production therefore requires large-deviation structure somewhere, and the anti-circularity contract forbids inserting it. From this measurement onward the campaign expected a disciplined negative, and said so in writing before the first sealed run.

III. A PUBLISHED CLASSICAL PAIR EVENT, REPLICATED

Before extending anything the campaign reproduced an existing result. Stueckelberg-Horwitz-Piron electro-dynamics evolves events under a parameter-dependent five-potential, so the mass shell is not fixed and $dt/d\tau$ may change sign along one worldline [3, 5]. Land's impulsive Coulomb scattering configuration was transcribed equation by equation with its conventions recorded, and its closed-form final velocity agrees with a numerical solution of the linear system it came from to 5.7×10^{-14} , its two published forms of the time component agree to 1.4×10^{-14} , and its Rutherford limit reproduces the published half-angle relation exactly [6]. The reversal threshold sits exactly at dimensionless coupling 2 as published.

The replication also produced a finding the campaign kept. With the source's own frozen-point convention the kick generator exponentiates to hyperbolic evolution that never reverses the time velocity, while the source's impulsive midpoint closure applies the same generator through a Cayley transform whose pole sits exactly at coupling 2. The published threshold is a property of the impulsive closure of the interaction rather than of continuous accumulation of the same force. Twenty-seven cells of the sweep have spacelike outgoing velocity, which is Stueckelberg's own requirement that a reversing worldline cross the spacelike region twice rather than a transcription error, and the committed record lists all of them. Below threshold the smoothed worldline has zero crossings of the observed time and above it exactly one, classified an annihilation fold by the sealed classifier, which is what ties the published pair event to the kinematics of Sec. II. The full analysis is a separate paper of this campaign [19] and its consequence for what follows is one clause. Any candidate whose reversals survive only because its interactions are closed impulsively is reproducing a Cayley pole and not a dynamical rate, so every claim-bearing run in this arc integrates continuously through every

interaction.

IV. THE SCHWINGER CHALLENGE AND ITS DISCIPLINED NEGATIVE

The first family capable of carrying a physical claim is a Stueckelberg-type system in which a hidden anharmonic coordinate is tilted by a Sauter slab,

$$H = \frac{1}{2}p_t^2 + \frac{1}{2}p_u^2 + \frac{1}{2}\omega_u^2 u^2 + \frac{\lambda}{4}u^4 + gEL \tanh(t/L) u, \quad (2)$$

with $\omega_u = 1.2$, $\lambda = 0.1$, $g = 0.25$, $L = 3$, thermal initialization in the shifted well at temperature 1, entry at $t = -4L$, and velocity Verlet at step 10^{-3} . A member reverses when p_t reaches zero, which is the fold of Eq. (1) in this model. The question is whether the fraction that reverses is organized by the Schwinger-shaped combination or by something else.

The pilot answered a prior question first. Across eight cells at 5×10^4 members each, with relative drift below 7×10^{-7} in every cell, the family has a critical field above which the deterministic crossing kick alone reverses the event, and below it reversal is fluctuation driven. The transition is razor sharp. At gap 0.5 the reversing fraction moves from 2.8×10^{-4} at field 0.65 to 0.9115 at field 0.80, three decades across a fifteen percent step in the field, and only two of the eighteen cells landed in the measurable band at all. Fixed rectangular grids therefore buy bounds rather than measurements, and every later manifest in this family is placed by bisecting a deterministic probe to a declared target.

Two variables were then declared and compared. Writing $d(P, E)$ for the deterministic minimum of the time momentum measured by a thermal-free probe, which is the effective gap the fluctuation must cover, and f for the reversing fraction, the two competing models are

$$-\log f = \alpha + \beta \left(\frac{d}{E} \right)^2 \quad \text{and} \quad -\log f = \alpha' + \beta' \frac{d^2}{E}, \quad (3)$$

the first a Gaussian tail in the effective gap and the second the shape the Schwinger exponent has. Both are fitted by weighted least squares on training gaps and asked to predict held-out gaps that are disjoint in the gap variable.

The first sealed attempt returned the third branch its own preregistration had defined. All twenty probe-placed cells landed in the measurable band, both prescription controls agreed per member exactly at $z = 0.00$ under fourth-order Runge-Kutta and under step halving, the zero-field null produced zero reversals, and the fitted effective-gap slope of 41.24 confirmed the pilot's two-point prediction. The effective-gap model beat the Schwinger-shaped model by a factor of 3.28 on held-out weighted error. But the declared bar required a held-out weighted mean squared error of at most 4 and the measurement was 81.22, because at that ensemble size the residual gap dependence of the suppression at fixed effective-gap variable is many binomial standard errors. The verdict

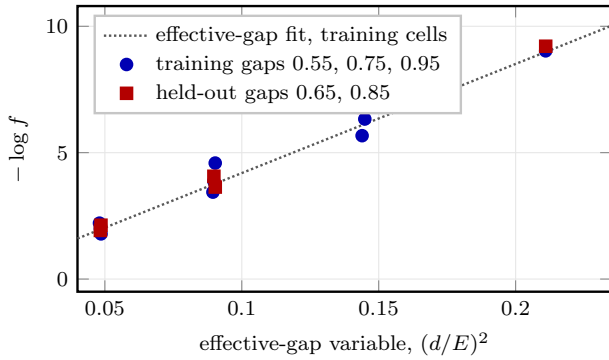


FIG. 1. The sealed axis comparison of PREREG-PF4-002. Each point is one probe-placed cell of 5×10^4 members, plotted at the declared effective-gap variable against the measured suppression. The dotted line is the weighted fit made on the training cells alone. The held-out cells lie on it, and the model's held-out weighted error is 0.38 of its own training error against a declared bar of 2, while the Schwinger-shaped variable of Eq. (3) is 3.14 times worse on the same held-out cells against the same bar. Ten of twelve training and seven of eight held-out cells met the sealed usability rule and are shown. All values are read from the committed evidence record.

was neither-axis, and the recorded reading is that the bars had been written in binomial units and were therefore testing a model nobody had claimed.

The second attempt declared in writing, before it existed, that it would set every bar against the model's own training inadequacy rather than against binomial noise, and it passed both. The effective-gap model's held-out weighted error is 0.3842 of its training error against a bar of 2, so it generalizes to gaps it never saw better than it fits the gaps it trained on, and the Schwinger-shaped model is 3.1398 times worse on the same held-out cells against the same bar. The fitted effective-gap slope is 43.18, ten of twelve training and seven of eight held-out cells were usable under the sealed rule, both prescription controls again agreed at $z = 0.00$, and the zero-field null again produced zero reversals. Figure 1 shows the comparison.

The label this earns is [demonstrated-in-model] and its content is narrow. In this declared family the suppression of time reversal is a Gaussian tail of the initial-condition measure in a measured effective gap, and the Schwinger-shaped variable fails to compete. That is a negative about a model family. It is not a statement about other families and it is not a statement about physical pair production. What it left open is the construction question, which became the summit of the whole arc. Does any constructible family have an exponent that is not a measure tail? The detailed treatment of this negative is a separate paper [20].

V. THE FAMILY BUILT TO ANSWER THE SUMMIT WAS EMPTY

A measure tail exists because there is a measure. The obvious way to remove it is to remove the ensemble, so the next family is a single deterministic event crossing an alternating train of Sauter slabs,

$$\text{tilt}(t) = gEL \sum_k (-1)^k \tanh\left(\frac{t - kD}{L}\right), \quad (4)$$

at field 0.6, slab length 3, spacing 18, integrated continuously by fourth-order Runge-Kutta. Each crossing deposits an analyticity-suppressed residual, the deposits accumulate, and the observable is the slab count at reversal. With one declared initial condition there is no distribution anywhere, so an exponential in this family could not be a measure tail.

The sealed run sealed itself unevaluable. Every one of the four training and three held-out velocities capped at 4000 slabs without a single reversal, and so did every cell of the mandatory planted control, so no fit existed to read and the record carries a manifest-design failure rather than a pass or a refutation. The diagnosis was that the manifest had been placed from a probe of a different observable at different constants, and the repair was a direct-observable probe run with the sealed dynamics verbatim. That probe measured the reversal count at twelve cells, three field strengths crossed with four velocities, and found zero reversals in all twelve at a cap of 300 slabs. The template registration that would have succeeded PF4-003 therefore cannot bind, because its own verification clause requires every manifest cell to have reversed below half the probe cap, and no cell reversed at all.

A separate study measured why, and its declared compound flag read False in a way that separates cleanly. The declared hypothesis had two halves, that the alternating slabs cancel the deposits and that the extrapolated slab count to reversal exceeds the cap. Cancellation is confirmed and it is strong. The net drift per slab is 3.3, 0.9, and 5.0 percent of the mean deposit magnitude in the three declared cells, so between 95 and 99 percent of every deposit is undone by the slabs that follow, with the sign of consecutive deposits flipping on 54, 68, and 27 percent of steps. The second half failed because it was written against the absolute drift, and the third cell's drift is positive at +0.0096 per slab, carrying the trajectory away from reversal rather than toward it. The corrected reading is stronger than the declaration. The pulse train does not merely deposit too slowly. Its residual after cancellation has no fixed sign, and at the largest declared field it accelerates the event away from the fold, which is why every probed cell is empty.

VI. TWO MANDATORY GATES, EACH OF WHICH HAD TO BE RUN TWICE

The campaign sequence makes conservation and covariance mandatory gates before any further claim, not optional follow-up analyses. Both gates were built, validated on exact controls, sealed, run, and passed. Both first runs then turned out to have certified nothing about the events they were about, and both had to be redeclared on probe-placed manifests.

The conservation instrument provides a complete census in which every trajectory ends in exactly one declared class, numerically failed, reversing, transmitted, or capped, so the missing-trajectory count is zero by construction and every reported statistic carries an explicit denominator with a flag that fires whenever a deletion would change a number. It tracks the conserved Hamiltonian every twenty-five steps for every member including failed members up to their failure step. And it declares the orientation charge to be the sign of $dt/d\tau$ before any claim run, with the conservation statement written as a path-degree rule, that the signed crossing count of any generic level equals the endpoint bookkeeping, which is exactly the statement that folds make orientation charge only in cancelling pairs. Its controls all pass. A harmonic control drifts by 4.4×10^{-7} against a control bar of 10^{-6} , a field-free cell classifies 2000 of 2000 transmitted, a deterministically reversing cell classifies 2000 of 2000 reversing, an absurd step size counts 200 of 200 numerically failed with zero missing, a short cap counts 100 of 100 capped, the analytic double fold obeys the path-degree rule at all 41 generic levels, and the deliberate deletion trap is mechanically flagged.

The first sealed gate run passed every bar and tested nothing. Its eight cells are all evaluable, every census sums exactly to 20000 with zero missing trajectories, the largest relative energy residual across cells is 6.5×10^{-7} against the sealed bar of 10^{-5} , and the path-degree rule holds at every generic level on all 96 declared polyline members with zero failures and zero refused levels. The same census also records zero reversing trajectories in every cell. All 160000 members transmitted. The sealed claim was about the accounting of apparent pair events and this grid contained none, so what the run certifies is the accounting machinery on transmitted worldlines. The named error is the manifest. A product grid of gaps against fields was declared where the sealed family's own cells had been placed by bisection, which is how the manifest landed outside the regime where reversals occur.

The repair binds cells from a committed placement probe that bisects the field at each declared gap until the reversing fraction lands inside a declared window, measuring 0.1000, 0.0970, 0.0995, and 0.1030 at the four bound cells, and it adds one bar that makes a vacuous pass impossible, that every evaluable cell must record a reversing count above zero and the run must record at least two thousand reversals in total. The rerun passes all of it. The four cells record 1998, 1952, 1881, and

2237 reversing members out of 20000 each, 8068 audited apparent pair events in total, every census exact with zero missing, the largest relative energy residual 7.3×10^{-7} against the same 10^{-5} bar, and zero path-degree failures and zero refused levels on all 48 declared polyline members. Conservation and complete accounting hold on cells that contain the events.

The covariance instrument audits five things, and four of them applied here. Translation covariance shifts the field profile and the initial data together and requires every event worldpoint to move by exactly the offset, measured at 2.7×10^{-13} on its validation members. Reparametrization invariance scales the Hamiltonian and requires the per-worldline event count to be exactly invariant while the rate per unit parameter scales by the declared factor, which it does exactly, and the witness this freezes is that the physical rate of this model class is per worldline and never per parameter tick. The boost audit runs on the family with genuine hyperbolic structure, the replication of Sec. III, where flow and boosts commute at 4.5×10^{-13} , the invariant is conserved at 7.1×10^{-11} , and the future-cone margin stays positive across the declared rapidity grid, so the never-reverses finding of that family is boost invariant. The observer audit measures the event count twice for a declared family of detector time functionals, once from the observer's own record and once from the model's detector prediction, and requires exact agreement at every declared observer. The gauge clause is recorded not applicable rather than passed, because the Sauter tilt is declared non-electromagnetic and has no field tensor, and the gauge machinery stands validated on a separate electromagnetic control at 2.2×10^{-13} for the physical history and 1.1×10^{-12} for the predicted canonical shift.

The first sealed covariance run returned vacuous, and its own anti-vacuity bar is what caught it. The declared deterministic member at each bound cell produces zero folds, so four cells were audited for the covariance of events that were not there. The named error is the member rather than the cell. The thermal ensemble reverses about ten percent of the time at these cells, so the members that reverse carry transverse momentum in the tail while the declared member did not. A committed member probe then scanned the initial transverse momentum at each cell and bound the members that fold, with verified fold counts 4, 2, 2, and 2. The rerun passes. Fold counts are exactly the probe's, 10 folds in total and every count even, translation shifts move the first-fold worldpoint by exactly the offset to 3.1×10^{-14} , the reparametrized rate ratio deviates from the declared factor by exactly zero, the boost audit repeats its values, and the observer count read from the observer's own record equals the detector-model prediction at every one of the six declared observers on every member. Observer-dependent particle number survives in this model, lawfully, because a declared detector model predicts every count exactly.

VII. THE MODEL CARRIES NO QUANTUM PHASE STRUCTURE

With both gates cleared the ceiling test is ungated. Its question is whether the fold model reproduces more than a classical worldline picture, and its discriminator was declared with the reasoning that makes it refutable. A quantum mode swept twice through an avoided crossing accumulates a relative phase between the two passages, so its excitation probability oscillates with the delay between them, which is Landau-Zener-Stueckelberg interferometry [11–14]. A classical ensemble of independent trajectories has no phase to accumulate. Both arms are reduced to one statistic, the largest nonzero-frequency power of the mean-removed discrete Fourier spectrum of the curve divided by its total power.

The quantum arm is a two-level mode with declared gap 0.30 and sweep rate 1, integrated by fourth-order Runge-Kutta, whose single-passage control must match the closed form

$$P = \exp\left(-\frac{\pi \Delta^2}{2v}\right) = 0.868167 \quad (5)$$

at the declared gap and sweep rate. The classical arm is the pilot fold dynamics with the single slab replaced by two Sauter pulses separated by the same delay, with the reversing fraction of a seeded thermal ensemble as its observable.

The first run failed two instrument bars and its apparent ceiling finding was vacuous. The single-passage control measured 0.8631 against Eq. (5), a deviation of 5.06×10^{-3} against a declared 2×10^{-3} , because the declared sweep range of forty natural units does not reach the asymptotic regime at that gap. The quantum oscillation statistic measured 0.2238 against a declared 0.30 while the recorded curve plainly oscillates, because the Stueckelberg phase between crossings at plus and minus a half-delay is the integral of the detuning between them and is therefore quadratic in the half-delay, so a grid uniform in the delay undersamples a chirp and spreads its power across bins. The norm of the two-level state is conserved to 5.5×10^{-14} , so the mode holds at most one excitation exactly. And the classical arm returned a reversing fraction of exactly 1.0 at every one of the twenty-four delays, all 2000 members of every cell reversing, so its oscillation statistic was computed on a constant curve and was zero for a reason with nothing to do with interference. The declared field was strong enough that two pulses reverse everything.

Three corrections followed, each to a named error. The sweep range became 120 natural units, the delay grid became uniform in the squared half-delay, and the classical field was bound from a committed probe that bisects it to a reversing fraction near one half and then verifies the fraction at the extreme delays. That probe rejected the originally declared delay range by its own window rule, because at the shortest delay the two pulse centres sit closer together than the pulse length and the pulses act

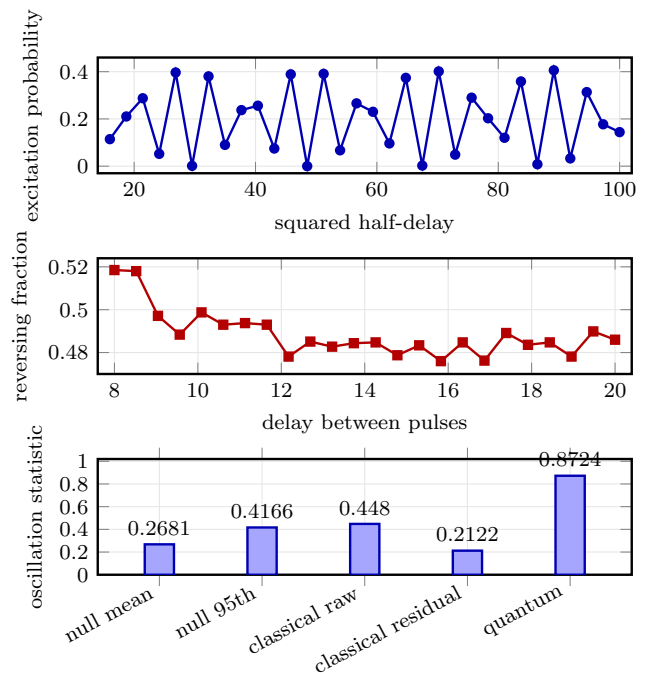


FIG. 2. The ceiling measurement after its four corrections. The top panel is the quantum arm’s excitation probability on the grid uniform in the squared half-delay, where the chirp becomes a single frequency. The middle panel is the classical arm’s reversing fraction at eight thousand members per delay, whose whole range is 0.0425 and which falls smoothly as the two pulses stop overlapping. The bottom panel places the declared statistic against the null distribution built from the same binomial counts, showing the null mean, the null ninety-fifth percentile, the raw classical value, the classical value after the declared smooth overlap model is removed from the curve and from every null curve alike, and the quantum value. The classical residual sits below the null mean at a p-value of 0.757 and the quantum value sits past every null draw. All values are read from committed evidence records.

as one stronger pulse, returning a fraction of exactly 1. A second probe measured 0.5245, 0.4945, 0.4750, 0.4835, and 0.4955 across the candidate range, all inside the declared window, and the corrected run bound that range.

The corrected run passed all three instrument bars. The single-passage control now measures within 1.49×10^{-3} of Eq. (5), the quantum oscillation statistic rose from 0.2238 to 0.8702 under the grid uniform in the squared half-delay, which confirms that the earlier failure was the grid rather than the physics, and the norm is conserved to 5.9×10^{-14} . The classical arm varies as the probes promised, between 0.4645 and 0.5245 across thirty-two delays. The ceiling finding then failed its own declared bar. The classical statistic measured 0.3626 against a declared 0.15 and the quantum-to-classical ratio 2.40 against a declared 3.

That bar was unattainable as declared, and the reason belongs to the statistic. On a thirty-two point curve the statistic is the largest of sixteen powers divided by their sum, which is a sizable fraction even for pure noise, and

each cell carries binomial noise of about 0.011 from two thousand members while the whole curve spans only 0.06. The next run measured that floor instead of asserting it, drawing twenty thousand null curves from binomial counts at the curve's own mean. The null mean is 0.2165 and its ninety-fifth percentile is 0.3310, so the declared bar of 0.15 sat below the statistic's floor and could not have been met by any curve of that length. Against that null the classical statistic has a p-value of 0.0262 and the quantum statistic a p-value of 0.0. The protocol had declared in advance that a small classical p-value would be escalated rather than read as a ceiling, and it was.

The escalation quadrupled the ensemble to eight thousand members per delay and tested a declared mechanism, that the hidden oscillator rings at its own period and meets the second pulse in or out of phase with that ringing. The structure survived the larger ensemble at a p-value of 0.02605, essentially unchanged from 0.0262 at a quarter of the members, so it is not shrinking the way a fluctuation would. The mechanism was refuted. The spectral peak sits at a period of 12.52 in delay while the declared ringing period is 5.236, and 12.52 is the span of the delay range, so the peak is the lowest nonzero frequency bin, which is where a monotone trend lands rather than an oscillation. The curve falls from a mean of 0.4942 over the first half of the range to 0.4829 over the second. Two pulses of length three separated by eight still overlap and by twenty do not, so the combined kick weakens smoothly with delay. The consequence recorded with it is that the declared statistic counts a trend as structure, so neither p-value was evidence about oscillation.

Removing a declared linear trend from the curve and from every null curve alike then failed its own first item and refuted the diagnosis that produced it. The classical statistic rose from 0.4480 to 0.5014 and its p-value fell from 0.02605 to 0.01185, while the quantum statistic was untouched at 0.8702 before and after, as predicted for a peak sitting at a high frequency. The trend itself is real, a drop of 0.0234 across the range against a standard error of 0.0056, which is 4.19 standard errors. The arithmetic of the rise is plain, since removing power from the lowest bin shrinks the total and a peak elsewhere takes a larger share. What the failure showed is that a linear model does not absorb an exponential approach to an asymptote, and that the null, which draws every cell at one common mean, counts any smooth departure from a straight line as structure.

The last refinement declared its model from the mechanism rather than fitting it blind. Two pulses of declared length three overlap by an amount falling exponentially with their separation, so the declared model is an asymptote plus an exponential approach at that length, linear in its two free coefficients, removed from the curve and from every null curve alike. It passed all three items and both findings hold. The classical statistic drops from 0.4480 to 0.2122 with a p-value of 0.757, which sits in the middle of the null rather than in its tail, the quantum statistic is 0.8724 after the same treatment, and the trend remains

real at 4.19 standard errors. Figure 2 shows the corrected picture.

The terminus was declared before that run so that a fourth refinement could not become a search for a wanted answer, and the reading it fixed is the one that holds. The classical arm carries exactly the smooth delay dependence its own mechanism predicts, and once that mechanism is accounted for there is nothing left in its spectrum that noise does not explain. The quantum arm interferes at a level no noise floor approaches. The fold model is capped as a classical representation. It does not carry the phase structure that produces interference, and this caps the model without touching the kinematic claim of Sec. II or any measured result of this arc.

VIII. DECAY IS EXCLUDED BY PARITY AND IMITATED BY BLINDNESS

A fold changes the observed branch count by exactly two. A same-species decay of one particle into two is a change of exactly one. Those two sentences settle the decay question for complete continuous worldlines under generic projections before any run, and the run measures whether the model obeys its own arithmetic and what a restricted observer sees instead.

The design had assumed the pair-creation configuration, in which the worldline begins and ends on one side of the observed level so the branch count is even. A committed probe measured that configuration absent on every bound member, since all four members end at their own largest observed time and begin at their own smallest, making both candidate windows exactly zero wide. The bound members are through-going, one branch in and one branch out, which is the configuration a decay question wants anyway. On a through-going worldline the complete observer's unsigned branch count is odd at every generic level, and a fold changes it by two, so a sweep reads one branch, then three, then one. The reframing was declared before the run. The same probe fixed the level ladder, because it measured the fold excursions to be as narrow as 0.000253 in observed time against a uniform ladder spacing near 0.2, so a uniform ladder steps over the three-branch fibers almost everywhere. The declared ladder merges a uniform background with nine levels inside each interval spanned by a consecutive pair of fold times.

The complete-observer arm passed with zero odd events. Four deterministic members were audited on 441 generic levels and 400 kept thermal polylines on five levels each. The complete observer's unsigned branch count took only the values 1 and 3 and every change between adjacent levels was minus two, zero, or plus two, giving zero odd events and zero parity failures on either arm, with fold counts 4, 2, 2, 2, all even. The frozen path-degree rule reported zero failures on all 441 deterministic levels and all 2000 thermal polyline checks, and the signed count on every member was exactly +1 at every level with spread zero, so no fold moved it. Folds make orientation charge

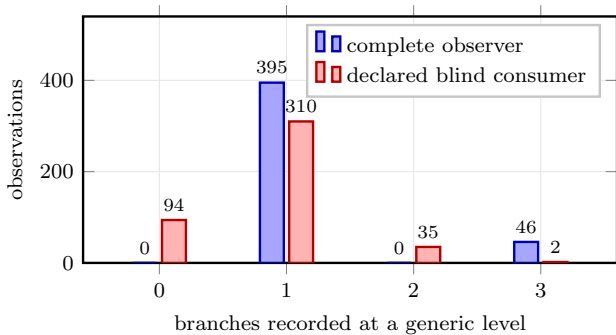


FIG. 3. What two observers of the same four worldlines record, pooled over all 441 generic levels of the declared ladder. The complete observer records only 1 or 3 branches, which is the parity fixed by the worldline endpoints, and every change between adjacent levels is minus two, zero, or plus two. The consumer declared blind to branches whose transverse coordinate lies in a fixed window records 0, 1, 2, or 3, so its parity is not conserved and 63 of its transitions are a rise by exactly one, which is what a decay of one particle into two looks like. Every one of its 441 counts was predicted exactly by its own detector model. All values are read from the committed evidence record.

only in cancelling pairs, measured rather than assumed.

The blind consumer is where the phenomenology appears. The declared consumer fails to record any branch whose transverse coordinate at the crossing lies in a closed window fixed from the probe record before the run, which the probe measured to hide 147 of 533 pooled crossings, 27.6 percent, some but not all. Across 441 observations that consumer failed to record 147 crossings and read a parity different from the complete observer on 129 of them, a flip fraction of 0.2925. Sixty-three of those were apparent decay events, meaning adjacent levels across which the blind count rose by exactly one, one particle becoming two with missing energy. Every single apparent count matched the detector-model prediction exactly, zero mismatches on 441 observations, computed by an independent route from the consumer's own record. On the member where all three branches of the zigzag lie inside the window the consumer sees nothing at all where the complete observer sees three. Figure 3 shows both observers. The anti-vacuity clause holds throughout, every member folding, the ladder resolving 46 multibranch levels, the consumer hiding 147 crossings, and the thermal census containing 1963 reversing members with the census summing exactly to 20000 and a largest relative energy residual of 6.3×10^{-7} .

Lifetimes were measured with an expected negative declared in advance. Real decay is memoryless, so the survival curve of a memoryless process has a coefficient of variation of exactly one. Re-running the frozen census at a ladder of step caps with the seed held fixed gives the exact cumulative distribution of the first-reversal time over the same members. Pooling the four cells, 1963 first-reversal times have mean proper time 18.92 and

standard deviation 3.01, a coefficient of variation of 0.159. Per cell it is 0.0526, 0.0530, 0.0694, and 0.0764, and the pooled figure is larger only because the four cells have different means. The largest deviation between the empirical survival curve and the exponential whose mean matches it is 0.506 pooled and near 0.57 in every cell, most of the available range, and shifting to the measured onset does not rescue it, giving a shifted coefficient of variation of 0.494 and a shifted deviation of 0.232. An exact cross-check on 32 first-reversal steps read from kept polylines gives mean 18.94 and coefficient of variation 0.188. First-reversal time in this family has a hard onset set by when the trajectory reaches the slab and a narrow spread after it, which is not a hazard at all, and it is the negative control's lesson again, that fold statistics are borne by the declared measure.

The answer the arc records is therefore two-sided. For the decay everyone means, no. A complete observer of a continuous worldline under a generic projection cannot see one particle become two, and that is parity rather than a limit of the search. For a decay nobody can distinguish from it without seeing every branch, yes. The parity-even zigzag of a parent plus a created pair reads as a decay with missing energy the moment one branch class goes unrecorded, at counts the consumer's own detector model predicts exactly. Apparent decay in this model is a measurement of consumer blindness and never a property of the worldline. This is the same structure the campaign's entropy track measured as an invisible leak [18], and it is the phenomenology of decays with unobserved products without any of the physics. The caveats belong beside it. This is four members and one declared blind window, so the counts of apparent decays are counts on one ladder and not a rate, the kept thermal polylines are half worldlines because the frozen census stops a member at its first reversal, and nothing in this section is sealed.

IX. THE SUMMIT QUESTION, ANSWERED UNDER SEAL FOR ONE OBSERVABLE

The open question left by the sealed negative was whether any constructible family has an exponent that is not a measure tail. A single deterministic crossing has one declared initial condition, so there is no distribution whose tail could produce a suppression, and an exploratory probe had already measured the per-crossing energy transfer of such a crossing to be exponential in the inverse velocity with a coefficient of determination of 0.9921, at a slope of -7.8186 against a naive estimate of -11.3097 . What that probe could not say is whether the exponent is dynamical or an accident of a fitted range.

The decisive property is analyticity. An exponent set by the field's nearest complex singularity is a dynamical quantity, because no measure tail knows where a function's poles are, and the adiabatic exponentiation of a two-level crossing is controlled exactly by that distance

[15, 16]. Writing d for the distance from the real axis to the nearest complex singularity of the force profile and P for the crossing velocity, the natural variable is the adiabaticity

$$\kappa = \frac{\omega_u d}{P}, \quad (6)$$

and in that variable the parameter-free estimate for the log transfer is a slope of exactly -2 . The observable throughout is the residual oscillator energy about the instantaneous well at exit, which is well defined for every profile tested including the ones with algebraic tails.

The first run declared its cells at fixed velocities and failed three of four bars. The timestep control passed at 6.9×10^{-11} , so nothing in it was numerical, and the sech-squared cell at width three reproduced the earlier probe's slope, -7.8154 against -7.8186 , an independent replication by a separate runner. The Lorentzian profile behaved as an analyticity-controlled exponent should, with coefficients of determination 0.99997, 0.99991, and 0.99933 and measured exponents 1.0095, 1.0344, and 1.0658 times the naive estimate, while the sech-squared profile gave 0.5430, 0.6910, and 0.8098 and drifted with width. The named error is the grid. Velocities were declared in absolute terms and applied to every width, so the three widths were compared in different dynamical regimes and the narrowest cells sat partly outside the adiabatic regime where the exponential form is asymptotic.

Placing each cell at a declared adiabaticity instead removed the drift entirely and refuted a declared expectation. The Lorentzian exponents in the variable of Eq. (6) are -2.0661 , -2.0639 , and -2.0607 across a factor of two in width, a spread of 0.26 percent, each sitting 3.3 percent above the parameter-free value, at coefficients of determination of 0.9998. The sech-squared exponents are -1.0490 , -1.0478 , and -1.0462 , a spread of 0.27 percent. The declared expectation that the sech-squared profile is not width-universal is refuted, and the earlier 39 percent spread was entirely an artifact of the grid. Both profiles carry an exponent fixed by the adiabaticity built from their own singularity distance. What they do not share is the constant, the Lorentzian sitting at the naive value and the sech-squared at 0.5245 of it. The run's verdict is nonetheless FAIL, on a bar requiring a coefficient of determination of 0.99 applied to both profiles when only one of them was predicted to be exactly exponential, and the sech-squared cells sit at 0.9886, 0.9887, and 0.9889.

Applying the gate content to the new profile then found a defect in the observable's definition before anything about it was sealed, which is what a gate exists to do. The conserved quantity of the extended system drifts by at most 8.6×10^{-14} , the census is complete with every crossing transmitted and no nonfinite member, and translation covariance is exact to 8.8×10^{-14} , so the transfer does not care where the field is centred. But moving the entry point from twenty widths to thirty changes the transfer by 1.289, 1.295, and 1.291 percent at the three widths against a bar of 10^{-8} . The diagnosis is that

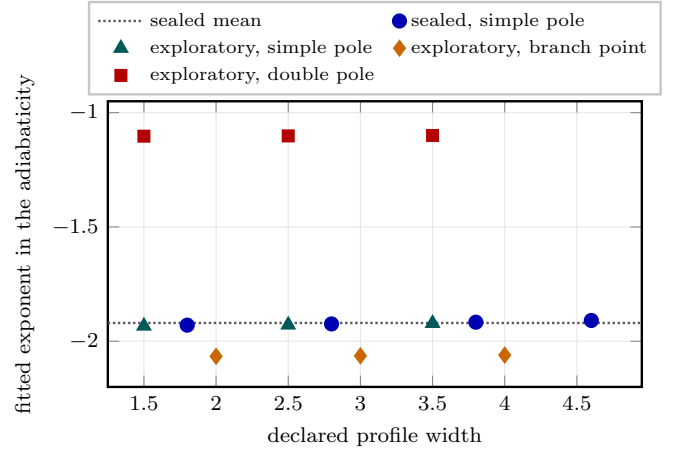


FIG. 4. Every fitted exponent of the summit sequence, plotted against the declared width of its field profile. The four sealed values span widths differing by a factor of 2.56 and agree within 1.07 percent of their mean, which the dotted line marks, against a declared bar of two percent, and that agreement is the sealed claim that the exponent depends only on the adiabaticity built from the singularity distance and not on the width. The two exploratory families whose nearest singularity is first order sit near the same value while the family whose nearest singularity is a double pole at the same distance sits at about 1.1, so distance alone does not fix the exponent. All values are read from committed evidence records.

the arctangent tilt approaches its asymptote algebraically rather than exponentially, so the field never fully turns off and a transfer measured from any finite entry carries a tail contribution. A second clause missed for a different reason, halving the timestep changing the transfer by about 3×10^{-7} against the same 10^{-8} , which is a bar set tighter than the declared timestep can deliver and is a fault in the declaration.

The span was then measured rather than assumed. Along a declared ladder of entry spans the Lorentzian transfer's successive relative changes are 1.278, 0.586, 0.130, and 0.015 percent, falling steadily, while the sech-squared transfer changes by at most 1.1×10^{-13} along the same ladder against a declared bar of 10^{-9} , so the effect belongs to algebraic tails and not to the method. The exponent is far more robust than the transfer, changing by at most 0.28 percent between the nearest and furthest spans, and the width universality survives at the furthest span with a spread of 0.36 percent. The tail contribution lands in the prefactor and leaves the exponent alone.

Separating the singularity's order from its tail behaviour was the last exploratory step, and it was declared with two mutually exclusive hypotheses so the run would discriminate rather than confirm. A Gudermannian tilt has a first-order pole at the same distance as the sech-squared family's double pole and has exponentially decaying tails, so the comparison holds the distance fixed and varies the order. The entry-point clause the Lorentzian profile failed at 1.3×10^{-2} is passed by this one at 5.7×10^{-11} , with

the timestep and translation clauses at 4.9×10^{-13} and 6.5×10^{-15} . The discriminator answered for the order. The Gudermannian exponents are -1.9333 , -1.9286 , and -1.9217 , a spread of 0.61 percent, while the sech-squared exponents on the identical grid are -1.1028 , -1.1015 , and -1.0997 , a spread of 0.29 percent, a ratio of 1.75. Two profiles whose nearest singularities sit at the same distance and differ only in order carry different exponents, so the distance alone does not fix it.

That run nonetheless failed on conservation, drifting at 1.9×10^{-10} against a bar of 10^{-10} , and the declared repair refuted itself. Halving the timestep left the drift at 2.0×10^{-10} , so the drift was not integration error. The diagnosis is a floating-point inconsistency in the tilt. A hyperbolic tangent of twenty rounds to exactly one in double precision while its own derivative is still nonzero at that point, so the tilt and the force become mutually inconsistent in the tails, and because the Hamiltonian carries the tilt the inconsistency reads as an apparent conservation violation. The sech-squared profile is immune because its force at the same point is far smaller. Evaluating the tilt so that it saturates together with its derivative dropped the worst drift to 2.6×10^{-12} and moved the exponents only in their seventh digit, exactly as the diagnosis predicted, and that run passes all eight of its items with entry-point independence at 3.0×10^{-9} , timestep independence at 6.6×10^{-13} , and translation covariance at 2.5×10^{-15} .

The sealed registration then bound the Gudermannian profile on a manifest disjoint from every prior run, four widths spanning a factor of 2.56 crossed with six declared adiabaticities, with every bar set at the precision the prior measurements support rather than at the precision they appeared to have. It passes all six. The four fitted exponents are -1.929735 , -1.924383 , -1.916823 , and -1.909268 , a spread of 1.07 percent against a declared bar of two percent. Coefficients of determination are 0.9989 or better in every cell. The extended Hamiltonian drifts by at most 3.7×10^{-13} against a bar of 10^{-11} , entry-point independence measures 9.2×10^{-9} against 10^{-7} , every transfer lies inside the declared signal window, and no cell is nonfinite and none folds. Figure 4 places those four values beside every exploratory exponent of the sequence.

The summit question has a sealed answer for one observable. A constructible family has a suppression whose exponent depends only on the adiabaticity built from the distance to the field's nearest complex singularity, and is therefore independent of the width. Each crossing is deterministic with one declared initial condition, so no distribution exists whose tail could produce the suppression. The exponent is dynamical and it is not a measure tail.

The registration's own non-claims bind that reading and are repeated here because they are the whole scope of the result. The observable is the per-crossing energy transfer and not an event rate, and the step between them remains blocked by the cancellation measured in Sec. V. The constant multiplying the adiabaticity is not claimed, and the exploratory observation that a first-order singularity

gives close to minus two while a second-order pole gives about -1.1 is not sealed. Nothing in this section is a claim about quantum field theory or about any physical process. The campaign also stopped here deliberately. Two runs had already corrected two declaration errors, and a third refinement aimed at a constant that was recognised after measurement rather than predicted before it would have been fitting the analysis to a wanted answer.

X. FIVE VACUOUS RUNS AND FOUR BARS THAT FAILED FOR THE DECLARATION

This arc contains the densest failure record in the campaign, and the record is load-bearing rather than decorative, so it is reported here as a sequence of facts with each design error named.

Five runs passed or reported on manifests containing none of the events their claims were about. The pulse-train registration of Sec. V sealed itself unevaluable because every cell capped without a reversal, and the direct-observable probe then measured the family empty at all twelve probed cells. The first conservation gate passed all four of its bars on a product grid whose eight cells recorded zero reversing trajectories, so its claim about the accounting of apparent pair events was untested by its own pass. The first covariance gate ran its audits on four deterministic members that produce no folds, and only its own anti-vacuity bar caught it. The first ceiling run computed its classical oscillation statistic on a curve that was exactly constant, because the declared field reversed every member at every delay. And the first summit run compared three widths in different dynamical regimes, so its cross-profile ratio test was contaminated and could not be read.

The rule that came out of them tightened three times and is now unconditional. After the empty family it required a manifest to be placed from a probe of the same observable at identical constants. After the event-free grid it applied to gate runs as well as to summit runs. After the fold-free members it required a probe verifying event presence for every bound cell and every bound member. And after the saturated classical arm, which was a new construction rather than an inherited manifest and so slipped past the rule's earlier wording, it became a requirement on any arm of any experiment. Every registration in this campaign must now cite a committed probe showing that its manifest contains the variation its claim is about.

Four further bars failed for reasons belonging to the declaration rather than to the model, and each is a different mistake. The first sealed axis comparison set its bars in binomial units, which tested a model nobody had claimed, and the correction declared in writing that its bars would be ratios to the model's own training inadequacy before that registration existed. The first ceiling bar of 0.15 sat below its own statistic's noise floor, which a later null measurement put at a mean of 0.2165 on a curve of that

length, so no curve could have met it. The gate on the Lorentzian profile set a timestep bar of 10^{-8} that the declared timestep could not deliver, missing at 3×10^{-7} for a reason with nothing to do with the profile. And a conservation bar of 10^{-10} was missed by a factor of two, its declared repair of halving the timestep was refuted by the rerun, and the drift turned out to be a floating-point inconsistency in the tilt, so the bar was eventually met by fixing an evaluation rather than by integrating harder.

Two more entries belong in the same record. A bar requiring a coefficient of determination of 0.99 was declared over two profiles when only one of them was predicted to be exactly exponential, and the other missed it at 0.9886, which is itself informative about the order of its singularity. And two runners in the summit sequence imported a fitting helper that regresses against the reciprocal of the adiabaticity rather than the adiabaticity itself, which produced positive exponents for a quantity already committed as negative. The error was found by comparison against a committed record, the runners were corrected, and the declared protocols were rerun with no bar and no design changed.

None of this is offered as evidence of rigor. It is offered because the sequence of verdicts is part of the evidence and because five of these failures would have been reported as successes by a pipeline that did not check whether its manifest contained its events.

XI. WHAT THIS PAPER ESTABLISHES

Tables I, II, and III register every run of the arc with its recorded verdict. Read together they answer the founding question inside its declared scope, and the answer has four parts.

The kinematic part is settled and is the only part that is a theorem. A fold of the observed time makes exactly two branches of opposite orientation and leaves the signed count alone. The instrument that finds them was validated against exact controls before any dynamics ran, its tolerances were derived from measured margins rather than chosen, and the folds it finds survive perturbation.

The dynamical part is a negative in the one family the campaign could populate and seal. Reversal suppression there is a Gaussian tail of the initial-condition measure in a measured effective gap, and the Schwinger-shaped variable does not compete. The family constructed to escape that verdict by removing the measure entirely turned out to contain no events, because its alternating slabs cancel almost all of every deposit and the residual has no fixed sign. Both mandatory gates hold on cells that contain events, conservation with every census exact and the orientation charge rule never failing, and covariance with every count invariant under the declared passive transformations and every observer-dependent count predicted exactly by a declared detector model.

The ceiling part is a measured limit. The model reproduces the kinematics of pair events and the phenomenol-

ogy of apparent decay under restricted observation, and it does not reproduce quantum phase structure. A two-level mode interferes at a level no noise floor approaches while the matched classical ensemble carries only the smooth delay dependence its own pulse overlap predicts. This caps the model as a classical representation and does not touch the kinematic statement or any other measured result here.

The fourth part is the one sealed positive and it is narrow. There exists a constructible family whose suppression exponent is fixed by the distance to the field's nearest complex singularity, independent of the profile's width across a factor of 2.56 to within 1.07 percent, measured on deterministic crossings where no distribution exists whose tail could produce it. That answers the construction question the sealed negative left open, for the per-crossing transfer and for nothing else. The step from a transfer to an event rate is blocked by the cancellation the mechanism study measured, and no claim about a pair production rate follows.

Three objects here do not depend on the substrates measured. The first is the parity argument of Sec. VIII, which excludes a same-species decay of one particle into two for complete continuous worldlines under generic projections by arithmetic rather than by search. The second is the lawful-observer clause, which allows observer-dependent particle number exactly when a declared detector model predicts every count, and which held on 441 observations with zero mismatches for a consumer that was blind by declaration. The third is the standing rule of Sec. X, which five failures in this arc paid for.

Nothing above is evidence that physical spacetime is a projection, that physical pair creation proceeds by folding, or that physical particle decay is a measurement artifact. Every dynamical statement here is about a declared finite model, and the paper's scope row says so once more.

XII. METHODS AND REPRODUCIBILITY

All runs follow the campaign's standing discipline. Model families, constants, grids, seeds, integrators, tolerances, estimators, stopping rules, and falsification bars were declared in the track documents before the runs they govern, and claim-bearing runs are governed by a preregistration whose file hash is recorded in an append-only seal ledger at its sealing commit, verifiable against the repository's git history. The nine sealed documents of this arc are the tolerance freeze, three registrations on the Schwinger question, two on conservation, two on covariance, and the sealed analyticity exponent. Verdicts are computed from the declared items by the governed runner rather than written afterward.

Every result-bearing run writes an append-only JSON evidence record containing the declared parameters, the measured outcomes, the computed verdict, the code commit, the dependency versions, and a canonical SHA-256 hash of the record body. The records for this paper are

TABLE I. The ledger of the instrument layer and the sealed Schwinger negative. Verdicts are recorded exactly as computed from the declared items of each run. Proved means a statement about smooth maps under its stated hypotheses. Measured in model means an exact measurement in a declared finite construction with append-only evidence records and bars, grids, seeds, and estimators fixed before the run. Key numbers are quoted from the committed records. Tables II and III carry the gates, the ceiling, the decay gate, the summit sequence, and the scope of the whole arc.

Run	Verdict	Key measurements	Label
Fold kinematics	proved	a nondegenerate critical point of the observed time changes the unsigned branch count by exactly two and the signed count by exactly zero, under the stated smoothness and genericity hypotheses	proved
PF0-FREEZE-001 instrument net	SEALED	branch counts 0, 1, 2 across the creation fold with signed count 0 at every slice, the reverse order across the annihilation fold, the cubic critical point classified degenerate and never a fold, the double fold giving band counts 1, 3, 3, 3, 1 with signed count +1 and both critical points at $\pm 1/\sqrt{3}$, separation exponent within 10^{-12} of one half, cross-language branch locations agreeing to 4.5×10^{-16} , classifier tolerances derived from a worst residual of 4.6×10^{-13} and a smallest fold curvature of 0.208	instrument
PF-1 structural stability	PASS	500 trials at each of twelve perturbation norms from 10^{-5} to 0.316, fold persistence exactly 1 through norm 0.123 and 0.956 at the largest with 22 of 500 lost, zero degenerate classifications anywhere, median separation exponent 0.5 throughout and within 2.8×10^{-6} of it at the largest perturbation	measured in model, exploratory
PF-2 negative control	measured	a single member gives exactly 12 folds with 7 positive and 6 negative orientation segments at relative drift 1.3×10^{-7} , and a 1000-member perturbed ensemble gives exactly 12 folds in every member, 12000 in total with zero dispersion, at mean relative drift 6.7×10^{-10} , with the same scenario reproducing its outcome hash bit for bit on a second substrate	measured in model, exploratory
PF-3 prior-art replication	replicated	the published closed form agrees with the linear system it came from to 5.7×10^{-14} and with its own published time component to 1.4×10^{-14} , the reversal threshold sits exactly at coupling 2, 27 sweep cells have spacelike outgoing velocity and are recorded, the smoothed worldline has zero observed-time crossings below threshold and exactly one above, classified an annihilation fold	replicated
PF-4 pilot	measured	18 cells at 5×10^4 members with drift uniformly near 6×10^{-7} , a razor-sharp transition from 2.8×10^{-4} at field 0.65 to 0.9115 at field 0.80 at gap 0.5, only 2 of 18 cells in the measurable band, so every later manifest in this family is probe-placed	measured in model, exploratory
PREREG-PF4-001 axis discrimination	neither-axis	all 20 probe-placed cells measurable, both prescription controls agreeing per member at $z = 0.00$, zero-field null exactly zero, fitted effective-gap slope 41.24, the effective-gap model beating the Schwinger-shaped model by 3.28 on held-out error, but held-out weighted error 81.22 against a bar of 4 because the bars were written in binomial units	measured in model, exploratory
PREREG-PF4-002 bars against model error	PASS	held-out weighted error 0.3842 of training error against a bar of 2, the Schwinger-shaped model 3.1398 times worse against the same bar, fitted slope 43.18, ten of twelve training and seven of eight held-out cells usable, controls at $z = 0.00$, zero-field null exactly zero	demonstrated in model

the instrument-net and ensemble records, the structural-stability pilot, the replication record, the pilot and the three sealed records of the Schwinger question, the direct probe and the mechanism study of the pulse-train family, the two conservation instrument and gate records with their placement probe, the two covariance instrument and gate records with their member probe, the six ceiling records with their two field probes, the decay gate with its blind probe, the eight records of the summit sequence, and the sealed analyticity record, all in **results/** in the campaign repository, each with its committed runner beside it. Failed and vacuous records are retained unchanged and each corrected declaration is a separate record, so the sequence of verdicts is itself part of the evidence. A reducer may summarize records but may not

rewrite them.

Every measured number and every figure coordinate in this paper is read from those committed records. The figure tables are generated deterministically by a committed script, `make_figdata_pf.py`, which uses only the Python standard library, has no free parameters, and asserts each value typed in this paper against the record value it came from before it writes anything, so a record that changed would break the build rather than disagree quietly with the text. Declared parameters quoted in the text, including the instrument-net controls, the frozen classifier tolerances, the seal ledger entries, and the bars of every registration, are checked the same way against the track documents and preregistrations that declared them.

TABLE II. The ledger of the empty family, the two mandatory gates, the ceiling sequence, and the decay gate. Both gates are recorded twice because both first runs met every bar on a manifest containing none of the events their claims were about. The ceiling sequence is recorded in full because four of its six runs corrected an error of the declaration. Key numbers are quoted from the committed records.

Run	Verdict	Key measurements	Label
PREREG-PF4-003 pulse train	unevaluable	every one of four training and three held-out cells capped at 4000 slabs without a reversal, and every cell of the mandatory planted control likewise, so no fit exists and the record carries a manifest-design failure	measured in model, exploratory
PF4-004 direct probe	empty	zero reversals at all twelve probed cells, three fields crossed with four velocities at identical constants, at a cap of 300 slabs, so the successor registration cannot bind under its own verification clause	measured in model, exploratory
PF4 mechanism study	flag False	net drift per slab 3.3, 0.9, and 5.0 percent of the mean deposit magnitude, so 95 to 99 percent of every deposit is cancelled, sign flips on 54, 68, and 27 percent of consecutive deposits, and the third cell's drift is +0.0096 per slab, carrying the event away from the fold	measured in model, exploratory
PREREG-PF5-001 conservation	PASS, content untested	all 8 cells evaluable, every census summing exactly to 20000 with zero missing, largest relative energy residual 6.5×10^{-7} against a bar of 10^{-5} , path-degree rule holding at every generic level on all 96 declared members with zero failures and zero refused levels, and zero reversing trajectories in every cell, all 160000 members transmitted	measured in model, exploratory
PREREG-PF5-002 probe-placed	PASS	1998, 1952, 1881, and 2237 reversing members out of 20000 each, 8068 audited apparent pair events, every census exact with zero missing, largest relative residual 7.3×10^{-7} against 10^{-5} , zero path-degree failures and zero refused levels on all 48 declared members	demonstrated in model
PREREG-PF6-001 covariance	vacuous	every declared deterministic member produced zero folds, so all four cells were audited for the covariance of events that were not present, and the run's own anti-vacuity bar recorded it vacuous rather than passing	measured in model, exploratory
PREREG-PF6-002 probe-placed	PASS	fold counts 4, 2, 2, 2 matching the member probe, 10 folds in total and every count even, first-fold worldpoints equivariant under the declared shifts to 3.1×10^{-14} , reparametrized rate ratio deviating from the declared factor by exactly 0, boost commutation 4.5×10^{-13} and invariant drift 7.1×10^{-11} with a positive future-cone margin, and the observer count equal to the detector-model prediction at all six declared observers on every member	demonstrated in model
PF-7 ceiling, first attempt	FAIL	single-passage control 5.06×10^{-3} from the closed form against a bar of 2×10^{-3} , quantum statistic 0.2238 against 0.30 because a grid uniform in delay undersamples a chirp quadratic in delay, norm conserved to 5.5×10^{-14} , and the classical arm returning exactly 1.0 at every one of 24 delays so its statistic was computed on a constant curve	measured in model, exploratory
PF-7b corrected grid and bound field	PASS, ceiling bar failed	single-passage control within 1.49×10^{-3} , quantum statistic 0.8702 under the grid uniform in the squared half-delay, norm 5.9×10^{-14} , classical fractions varying from 0.4645 to 0.5245, but the classical statistic 0.3626 against a declared 0.15 and the ratio 2.40 against 3	measured in model, exploratory
PF-7c noise floor	PASS	the null floor measured rather than asserted exactly as the protocol required	measured in model, exploratory
PF-7d ringing test	PASS, mechanism refuted	ninety-fifth percentile 0.3310 over 20000 draws, so the declared 0.15 was unattainable, classical p-value 0.0262 and quantum p-value 0.0, and the ceiling reading withheld and escalated exactly as the protocol required	measured in model, exploratory
PF-7e linear detrend	FAIL	structure survives four times the ensemble at p-value 0.02605 against 0.0262 at a quarter of the members, but the spectral peak sits at period 12.52 against a declared ringing period of 5.236, which is the lowest nonzero bin and therefore a trend, the curve falling from 0.4942 to 0.4829 across the range	measured in model, exploratory
PF-7f declared smooth model	PASS	removing a declared linear trend raised the classical statistic from 0.4480 to 0.5014 and lowered its p-value from 0.02605 to 0.01185, refuting the diagnosis, while the trend is real at 4.19 standard errors and the quantum statistic is untouched at 0.8702	measured in model, exploratory
PF-8 decay gate	PASS	removing an asymptote plus an exponential approach at the declared pulse length drops the classical statistic from 0.4480 to 0.2122 at a p-value of 0.757, the quantum statistic measures 0.8724 after the same treatment, and the trend remains real at 4.19 standard errors	measured in model, exploratory
		441 generic levels on four members and 400 thermal polylines, complete-observer counts only 1 and 3 with every change minus two, zero, or plus two, zero odd events and zero parity failures, zero path-degree failures on 441 deterministic and 2000 thermal checks with signed count exactly +1 and spread 0, the blind consumer hiding 147 of 533 crossings, flipping parity on 129 of 441 observations, producing 63 apparent decays, and matching its own detector model on all 441 with zero mismatches, and lifetimes with coefficient of variation 0.159 against the 1 a memoryless process requires	measured in model, exploratory

The instrument layer runs in two independent implementations, one in MATLAB with event-located adaptive integration and one in Python with fixed-step symplectic integration, and the frozen tolerances bind both paths separately, with an adaptive path at relative tolerance 10^{-10} and a fixed-step path at step 10^{-3} valid to evolution parameter 40 [17]. Cross-substrate determinism is part of the freeze, and the same scenario on a workstation and in a container reproduces both its scenario hash and its outcome hash bit for bit. The Sauter-family ensembles are brute-force with no importance sampling and no adap-

tive seeding, with seeds derived from declared formulas in the governed runners, and every trajectory is counted in every denominator including the numerically failed, the capped, and the transmitted. The summit sequence integrates single deterministic crossings by fourth-order Runge-Kutta at declared steps of 2×10^{-4} and 10^{-4} , with the extended two-degree-of-freedom Hamiltonian's drift, the entry-point independence, the timestep independence, and the translation covariance all measured per run rather than assumed. All runs executed on a single workstation and require no special hardware.

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- [1] H. Whitney, On singularities of mappings of Euclidean spaces I. Mappings of the plane into the plane, *Ann. Math.* **62**, 374 (1955).
 - [2] V. I. Arnold, S. M. Gusein-Zade, and A. N. Varchenko, *Singularities of Differentiable Maps, Volume 1* (Birkhäuser, Boston, 1985).
 - [3] E. C. G. Stueckelberg, La signification du temps propre en mécanique ondulatoire, *Helv. Phys. Acta* **14**, 588 (1941).
 - [4] R. P. Feynman, The theory of positrons, *Phys. Rev.* **76**, 749 (1949).
 - [5] L. P. Horwitz and C. Piron, Relativistic dynamics, *Helv. Phys. Acta* **46**, 316 (1973).
 - [6] M. Land, Pair production in classical Stueckelberg-Horwitz-Piron electrodynamics, *J. Phys.: Conf. Ser.* **845**, 012025 (2017), arXiv:1604.01625.
 - [7] F. Sauter, Über das Verhalten eines Elektrons im homogenen elektrischen Feld nach der relativistischen Theorie Diracs, *Z. Phys.* **69**, 742 (1931).
 - [8] W. Heisenberg and H. Euler, Folgerungen aus der Diracschen Theorie des Positrons, *Z. Phys.* **98**, 714 (1936).
 - [9] J. Schwinger, On gauge invariance and vacuum polarization, *Phys. Rev.* **82**, 664 (1951).
 - [10] G. V. Dunne, Heisenberg-Euler effective Lagrangians: Basics and extensions, in *From Fields to Strings: Circumnavigating Theoretical Physics* (World Scientific, Singapore, 2005), arXiv:hep-th/0406216.
 - [11] L. D. Landau, Zur Theorie der Energieübertragung II, *Phys. Z. Sowjetunion* **2**, 46 (1932).
 - [12] C. Zener, Non-adiabatic crossing of energy levels, *Proc. R. Soc. London A* **137**, 696 (1932).
 - [13] E. C. G. Stueckelberg, Theorie der unelastischen Stöße zwischen Atomen, *Helv. Phys. Acta* **5**, 369 (1932).
 - [14] S. N. Shevchenko, S. Ashhab, and F. Nori, Landau-Zener-Stückelberg interferometry, *Phys. Rep.* **492**, 1 (2010).
 - [15] A. M. Dykhne, Adiabatic perturbation of discrete spectrum states, *Sov. Phys. JETP* **14**, 941 (1962).
 - [16] J. P. Davis and P. Pechukas, Nonadiabatic transitions induced by a time-dependent Hamiltonian in the semi-classical/adiabatic limit: The two-state case, *J. Chem. Phys.* **64**, 3129 (1976).
 - [17] E. Hairer, C. Lubich, and G. Wanner, *Geometric Numerical Integration*, 2nd ed. (Springer, Berlin, 2006).
 - [18] A. H. Bond, Entropy across singular projections: Fiber multiplicity, caustics, and observer-relative information, Zenodo (2026), doi:10.5281/zenodo.21789011.
 - [19] A. H. Bond, The pair-creation threshold of impulsive Stueckelberg-Horwitz-Piron scattering is a Cayley-transform pole, Zenodo (2026), doi:10.5281/zenodo.21790096.
 - [20] A. H. Bond, A sealed negative for Schwinger scaling in a resolved-field hidden-dynamics family, Zenodo (2026), doi:10.5281/zenodo.21798545.
 - [21] A. H. Bond, Games and decisions under projection: Blackwell wedges, budget flips, and prospect signatures from declared consumers, Zenodo (2026), doi:10.5281/zenodo.21818268.
 - [22] A. H. Bond, Five audits under one consumer discipline: Renormalization, quantum Darwinism, lattice hydrodynamics, fluctuation theorems, and gauge redundancy in exact finite models, Zenodo (2026), doi:10.5281/zenodo.21821378.
 - [23] A. H. Bond, Engineered observation limits in exact finite models, Zenodo (2026), doi:10.5281/zenodo.21833690.

TABLE III. The ledger of the summit sequence and the scope of the whole arc. The exploratory runs are kept because three of them failed and each failure named the error that the next run corrected. The last row is a non-claim and governs every row of this table and of Tables I and II.

Run	Verdict	Key measurements	Label
PF4-005 analyticity, velocity grid	FAIL	timestep control 6.9×10^{-11} , the sech-squared cell at width three replicating the earlier probe's slope at -7.8154 against -7.8186 , Lorentzian coefficients of determination 0.99997, 0.99991, and 0.99933 with exponent ratios 1.0095, 1.0344, and 1.0658 against sech-squared ratios 0.5430, 0.6910, and 0.8098, but the velocity grid compared three widths in different dynamical regimes	measured in model, exploratory
PF4-005b adiabaticity grid	FAIL	Lorentzian exponents -2.0661 , -2.0639 , and -2.0607 , a spread of 0.26 percent, 3.3 percent above the parameter-free value, at coefficients of determination 0.9998, sech-squared exponents -1.0490 , -1.0478 , and -1.0462 , a spread of 0.27 percent, refuting the declared expectation, timestep control 1.8×10^{-13} , and the failed bar a coefficient of determination of 0.99 declared over both profiles when only one was predicted to be exactly exponential	measured in model, exploratory
PF4-006 gate content	FAIL	conserved quantity drifting at 8.6×10^{-14} , complete census with every crossing transmitted, translation covariance 8.8×10^{-14} , but moving the entry point from twenty widths to thirty changing the transfer by 1.289, 1.295, and 1.291 percent against a bar of 10^{-8} because the tilt approaches its asymptote algebraically, and a timestep bar the declared timestep could not deliver, missing at about 3×10^{-7}	measured in model, exploratory
PF4-007 span ladder	PASS	Lorentzian successive relative changes 1.278, 0.586, 0.130, and 0.015 percent along the declared span ladder while the sech-squared transfer changes by at most 1.1×10^{-13} against a declared bar of 10^{-9} , the exponent changing by at most 0.28 percent between the nearest and furthest spans, and width universality surviving at the furthest span with a spread of 0.36 percent	measured in model, exploratory
PF4-008 pole order	FAIL	the entry clause the Lorentzian failed at 1.3×10^{-2} passed at 5.7×10^{-11} , Gudermannian exponents -1.9333 , -1.9286 , and -1.9217 at a spread of 0.61 percent against sech-squared exponents -1.1028 , -1.1015 , and -1.0997 on the identical grid, a ratio of 1.75 refuting the distance-only hypothesis, but conservation drifting at 1.9×10^{-10} against a bar of 10^{-10}	measured in model, exploratory
PF4-008b halved timestep	FAIL	the declared repair refuted itself, the drift measuring 2.0×10^{-10} at half the step, so the drift was not integration error and the repair was wrong	measured in model, exploratory
PF4-008c corrected tilt evaluation	PASS	evaluating the tilt so that it saturates together with its derivative dropped the worst drift to 2.6×10^{-12} and moved the exponents only in their seventh digit, with entry-point independence 3.0×10^{-9} , timestep independence 6.6×10^{-13} , and translation covariance 2.5×10^{-15} , and all eight items passing	measured in model, exploratory
PREREG-PF4-009 sealed exponent	PASS	four fitted exponents -1.929735 , -1.924383 , -1.916823 , and -1.909268 across widths spanning a factor of 2.56, a spread of 1.07 percent against a declared bar of two percent, coefficients of determination 0.9989 or better in every cell, Hamiltonian drift at most 3.7×10^{-13} against 10^{-11} , entry-point independence 9.2×10^{-9} against 10^{-7} , every transfer inside the declared window, and no cell nonfinite and none folding	demonstrated in model
Scope	non-claim	the kinematic statement is a theorem and everything else here is measured in declared finite models at declared constants and a declared step, the sealed observable is a per-crossing energy transfer and not an event rate, the constant multiplying the adiabaticity is not claimed, apparent decay here is a measurement of consumer blindness and never a property of the worldline, and no result in this paper is evidence that physical pair creation or physical particle decay works this way	non-claim